

SHAPPROP: SCM-guided synthetic augmentation with composite TPE search and SHAP-rule feasibility for glycemic tabular models

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Abstract

Introduction: Gradient-boosted decision trees on observational clinical tables often learn shortcuts that contradict physiology. Post-hoc explainers such as SHAP summarise the fitted mapping but do not, by themselves, encode domain structure into training data.

Methods: We implement SHAPPROP for free-living CGM (DINAMO) in `scripts/main.py`. A compact list of signed *structural causal model* (SCM) rules (`d1namo.SCM_RULES`) defines (i) a directed acyclic order over inputs via feature→feature edges and (ii) optional target-directed edges into *future glucose*. On min–max scaled columns, we fit ordinary least squares (OLS) structural modules child|parents from the current (possibly augmented) training pool, estimate per-child residual scales for Gaussian shocks, and draw synthetic rows in topological order. The synthetic label combines a linear head on the row, an Optuna-sampled residual band, and a *rule push* that shifts y along the signed sum of target rules evaluated at the row. An optional insulin *anchor* breaks meal–insulin co-administration confounding when insulin targets glucose. Each TPE trial proposes noise, push strength, residuals, and replay parameters, materialises a batch of synthetic rows, retrains default LightGBM (squared error, early stopping on

real validation rows), and maximises a *composite* score in $[0, 1]$ blending validation RMSE quality relative to a baseline run with full feasibility weighting. Outer iterations refit the SCM on augmented data, append an accepted batch, and stop when neither composite nor feasibility clears a margin. Feasibility is a weighted mean over rules: for target edges we score per-patient SHAP values (mass- and count-based agreement with the prior sign); for feature–feature edges we use SHAP interaction blocks.

Results: We export `results/dinamo_shap_feasibility.json`, an SCM graph `.dot`, and before/after explanation figures; numeric cells in Table ?? should be refreshed after `python scripts/main.py`.

Discussion: Training stays on supported LightGBM losses; priors enter through SCM-consistent synthetic rows and a post-fit feasibility functional, which keeps auditing simple but couples scores to the validation sample and rule sparsity.

Keywords: Explainable AI, LightGBM, Synthetic data, Structural causal models, Optuna, TPE, SHAP, Domain knowledge, Biomedical machine learning

1. Introduction

Motivation and problem setting

Tabular predictors are routinely deployed in clinical decision support, operational forecasting and cohort analytics [? ?]. Labels and inputs are almost always observational, reflecting care pathways, documentation and selection rather than a controlled experiment [?]. Flexible models therefore learn whatever regularities compress the table, including *shortcuts* that need not survive changes in protocol, device, or population [?]. Standard supervised learning does not, by itself, import the stable structure of physiology or pharmacology.

This matters in two ways. First, **generalization**: performance degrades when deployment covariates drift from training [? ?]. Second, **interpretability**: popular post-hoc explainers summarise the fitted mapping, not whether that mapping is compatible with external knowledge [? ? ?]. SHAP values [?] are faithful to the model’s cooperative-game definition but can still mislead reviewers when the model’s partial associations disagree with textbook mechanisms [?].

Related work

A long line of work argues that explainability should be grounded in causal or domain structure [? ? ? ?]; recent surveys summarise prediction, generation, and explanation on tabular data [? ?]. SHAPPROP stays lightweight: no LLM inside the training loop, no custom LightGBM objective—only squared error on scaled targets, SCM-structured synthetic rows, and SHAP-rule feasibility scored after each trial.

Contributions

(1) An SCM-guided sampler that fits linear structural modules from data, injects calibrated shocks in topological order, and applies a signed *rule push* on the synthetic label from target-directed rules. (2) A single-branch Optuna TPE loop that maximises a bounded composite of validation RMSE quality and rule feasibility, with outer iterations that refit the SCM on augmented training data. (3) A feasibility functional that mixes per-row SHAP agreement for target rules with interaction-matrix scores for feature–feature rules, exported alongside graph and explanation diagnostics.

2. Methods

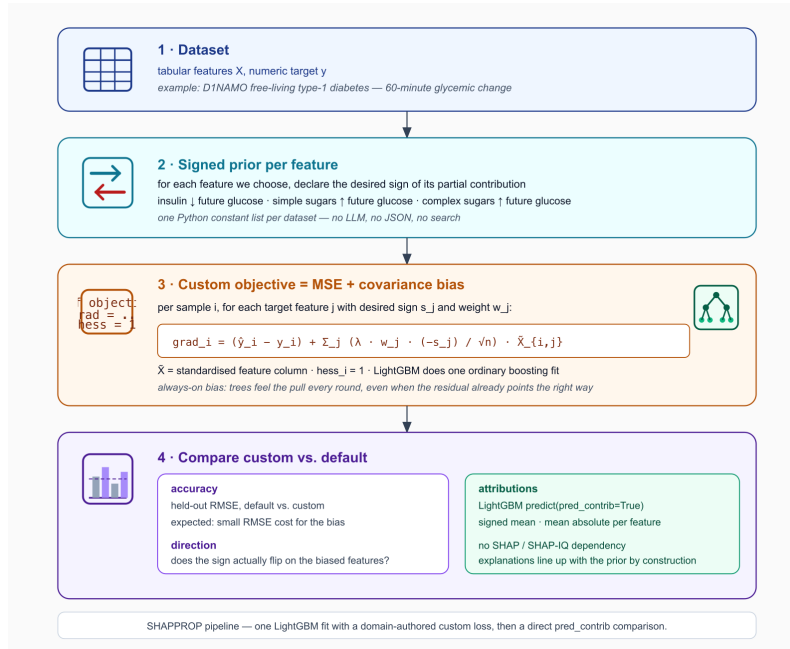


Figure 1: High-level orchestration: standard LightGBM regression on scaled targets with early stopping on real validation rows; Optuna TPE proposes SCM sampling and label-push parameters; feasibility is evaluated post-hoc from SHAP against declared rules. Source vector: `images/shapprop_algorithm_flowchart.svg`.

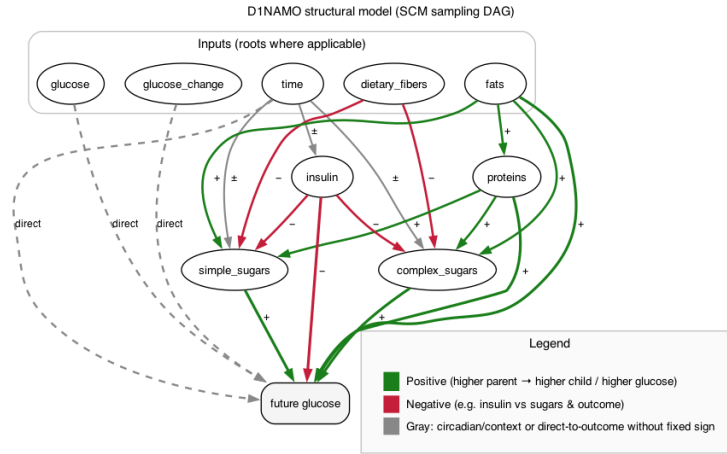
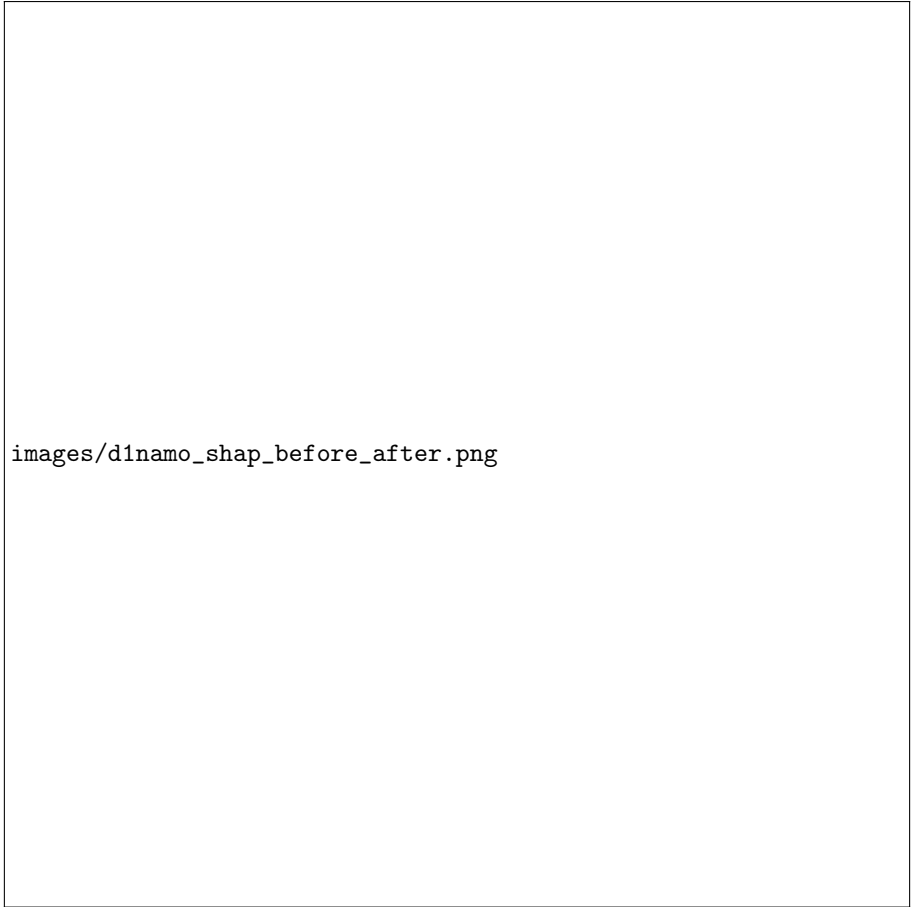


Figure 2: D1NAMO SCM prior: `d1namo.SCM_RULES` lists weighted directed edges (feature→feature structure and optional edges into *future glucose*). `export_scm_dag_dot` in `scripts/visuals.py` writes `images/d1namo_scm_graph.dot`; render to PNG for the manuscript.



images/d1namo_shap_before_after.png

Figure 3: Validation explanations before vs. after augmentation: SHAP bee-swarm columns and a ShapIQ second-order interaction graph on a fixed validation subsample (`scripts/visuals.py`).

2.1. Pipeline

`scripts/d1namo.py` fixes `NAME`, `TASK`, `TARGET`, `CSV`, `FEATURE_COLUMNS`, and `SCM_RULES`. `scripts/lib.py` supplies graph utilities (`scm_topological_order`, `scm_structural_parents`), TreeSHAP main and interaction values, and feasibility aggregation. `scripts/main.py` loads the table, min–max scales inputs and target on the training split, holds out a test fraction, and splits the remaining pool into train vs.

validation (validation is never synthetically augmented). It fits a baseline regressor, then runs the outer augmentation loop in §??.

2.2. DINAMO: SCM-guided synthetic augmentation

Let $\mathcal{G}$ be the rule list visualised in Fig. ??. Edges with end $\neq y$ induce structural parents for child features; Kahn ordering with a fixed column tie-break yields a generation order. For each child c with parent indices $P(c)$, we fit $\hat{x}_c = \beta_c^\top x_{P(c)} + b_c$ by OLS on the current augmented training matrix X in $[0, 1]^d$. Residual standard deviations σ_c supply structural noise scales. For a trial we initialise a row, fill exogenous or parentless coordinates by copying stratified real rows, then recursively set

$$x_c \leftarrow \text{clip}(\hat{x}_c + \nu \sigma_c z_c, 0, 1), \quad z_c \sim \mathcal{N}(0, 1),$$

with global shock multiplier ν proposed by TPE. When the rules include $\text{insulin} \rightarrow y$, an optional *insulin anchor* replaces x_{insulin} to mimic counterfactual dosing independent of meal-driven variation. The scaled label is

$$y \leftarrow \text{clip}(h(x) + \delta + \rho \cdot p(x), 0, 1),$$

where h is an OLS linear head $x \mapsto y$ fit on the augmented pool, δ is a trial-sampled band around the head, ρ is a trial-sampled push magnitude (bounded), and $p(x) = \sum_{(u \rightarrow y)} e_u (x_u - \frac{1}{2})$ sums target-directed rules (u, y) with signed weights e_u . Each trial replays its parameters to draw a full batch, concatenates it to training, retrains LightGBM, and returns the composite score below. An outer iteration accepts the batch if composite improves against the previous iteration *or* feasibility improves by a fixed margin; then the SCM is refit on the enlarged pool.

2.3. Composite TPE objective

Let RMSE_0 be validation RMSE after the baseline fit on real training rows only. For any candidate model with validation RMSE r and feasibility $f \in [0, 1]$, define RMSE channel quality $q_{\text{rmse}} = \text{clip}(\frac{1}{2} + \frac{1}{2}(\text{RMSE}_0 - r)/\text{RMSE}_0, 0, 1)$ and composite $S = (1 - w) q_{\text{rmse}} + w f$ with $w \in [0, 1]$ (the implementation defaults to $w=1$, emphasising rule alignment while still regularising via q_{rmse}). TPE maximises S ; subsampled rows accelerate feasibility inside trials, while baseline and final reporting use fuller validation statistics.

2.4. Feasibility scoring

Each rule carries a weight $w \in [0, 1]$ and a sign. For a target-directed rule on feature u , we take the vector of per-row SHAP contributions $\phi_{i,u}$ and compute a geometric mean of (i) the fraction of absolute SHAP mass with $\text{sign}(\phi_{i,u}) = \text{sign}(e_u)$ and (ii) the fraction of sufficiently non-zero rows with the same sign agreement. For a structural rule $u \rightarrow v$, we apply the same construction to the off-diagonal SHAP interaction values $I_{i,u,v}$. The overall feasibility is the weighted mean of per-rule scores; when TreeSHAP interactions are unavailable the implementation falls back to mean `pred_contrib` ratios for target edges only.

2.5. Data and domain priors

DINAMO (regression).. Free-living CGM with self-reported meals and insulin; target *future glucose*. After dropping missing rows, $n \approx 6731$ with nine inputs: glucose, glucose_change, time, macronutrients (simple_sugars, complex_sugars, proteins, fats, dietary_fibers), and insulin. The active `SCM_RULES` set in the repository

is intentionally small (e.g. a dominant insulin→*future glucose* constraint) so that feasibility focuses on clinically salient directions; additional edges can be re-enabled in `scripts/dinamo.py`.

2.6. Evaluation protocol

Train/test split: 75/25 on the full cohort; within the training pool, 50% is reserved as validation for early stopping and feasibility (never augmented). LightGBM uses fixed shallow defaults with early stopping rounds on validation. We report validation and test RMSE in original glucose units, composite score, feasibility, and exported SHAP diagnostics.

3. Results

Table ?? lists baseline vs. augmented metrics on the fixed splits produced by `scripts/main.py`.

Table 1: D1NAMO run summary (fill from console output / `results/dinamo_shap_feasibility.json` after `python scripts/main.py`).

Setting	Val RMSE	Val feas.	Composite	Test RMSE
Baseline (real train only)	—	—	—	—
+ SCM synthetic batches	—	—	—	—

D1NAMO.. The implementation matches §?? and Figs. ??–??. per-rule breakdowns appear under `rules` in the exported JSON.

4. Discussion

SHAPPROP separates **(i)** a sparse signed SCM prior, **(ii)** data generation that respects fitted linear structure plus optional counterfactual anchors, and **(iii)** feasibility as a post-fit functional of SHAP values aligned to those rules. Training uses only the standard squared-error booster path. Limitations include sensitivity of interaction-based scores to validation size, bias–variance trade-offs when $w=1$ on feasibility, and the fact that OLS structural modules on observational data need not identify a causal mechanism even when they improve interpretability metrics.

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